

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250278399

Kind Code

A1

Publication Date

September 04, 2025

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SYSTEMS AND METHODS FOR OVER-TIME EVENT MONITORING WITH ENHANCED RULES CONFLICT CHECKING WITH DATA VALIDATION

Abstract

A computer system for performing over-time event monitoring is provided. The computer system includes at least one processor in communication with at least one memory device. The at least one processor is programmed to scan a plurality of data messages and detect a first data message in the plurality of data messages. The first data message is associated with a first event that includes the first data message and a second data message. The at least one processor is also programmed to scan the plurality of data messages for the second data message associated with the first event. If the second data message is not detected based on the scan, the at least one processor is also programmed to determine whether the first event has expired. If the first event has expired, the at least one processor is programmed to transmit an error message to a user computer device.

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Family ID: 95942015

Appl. No.: 19/209452

Filed: May 15, 2025

Related U.S. Application Data

parent US continuation 17079088 20201023 parent-grant-document US 12326851 child US 19209452

Publication Classification

Int. Cl.: G06F16/23 (20190101); G06F11/30 (20060101); G06F16/25 (20190101)

U.S. Cl.:

CPC G06F16/2365 (20190101); G06F11/3041 (20130101); G06F16/2358 (20190101); G06F16/258 (20190101);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of and claims priority to U.S. patent application Ser. No. 17/079,088, filed Oct. 23, 2020, which is hereby incorporated by reference herein in its entirety.

BACKGROUND

[0002] The field of the invention relates generally to data validation, and more specifically to system and methods for providing over-time event monitoring with enhanced rules conflict check with data validation.

[0003] In many cases, computer devices provide requested or required information to other computer system or applications. Each of these applications may require the different data or that the data is required to be in a specific format. When a computer device, such as one associated with a database, had to provide data to multiple different systems, the different systems may require the data in different formats, or require different data to be provided for similar requests. Inconsistent data validations can allow undetected data errors to be transmitted to other computer systems, where the data errors can negatively impact the other computer systems. In some situations, the data injection point may be unknown or difficult to find. This increases the difficulty of and the time to find the point where the error is introduced and fix the error. This can also increase the complexity of fixing the error as multiple computer systems or applications may rely on the data associated with the data error. Accordingly, it would be advantageous ensure that the data is properly formatted and provided for each requesting system or application.

[0004] Furthermore, each message can represent an individual event or portion of an event. In addition, an event may take place over a period of time and may include multiple messages to complete the event. However, it can be difficult to recognize when an event is not completed due to missing one or more required messages, especially when the event takes place over a significant period of time.

BRIEF DESCRIPTION

[0005] The present embodiments may relate to systems and methods for performing enhanced rules conflict checking. A data integrity and validation (DIV) system, as described herein, may include a data integrity and validation (“DIV”) computer device. The DIV computer device may be configured to i) store a plurality of rules for transmitting to a plurality of destination systems, wherein the plurality of destination systems includes a first destination system and a second destination system; ii) receive, from a requesting system, a data message for transmission to the plurality of destination systems, wherein the data message is configured to be routed to each of the plurality of destination systems, wherein the data message includes a first plurality of fields for the first destination system and a second plurality of fields for the second destination system, wherein the data message is assembled by the requesting system from a source database based on a predetermined data format including a plurality of fields to be populated from the source database; iii) compare the data message to the plurality of rules to validate the data message; iv) if the data

message is validated, instruct the requesting system to transmit the data message to the plurality of destination systems; v) determine one or more validation errors based on the comparison; vi) determine whether or not to validate the data message based on the one or more validation errors, wherein the one or more validation errors include at least one of missing data, incorrect data, improperly formatted data, and failed conditional logic; vii) determine an error level for each validation error of the one or more validation errors, wherein the error level includes at least one of a low level and a high level; viii) validate the data message if none of the one or more validation errors includes a high level; ix) indicate that the validation failed if at least one validation error of the one or more validation errors includes a high level; x) request a user input in response to a validation error including a high level; xi) adjust the data message to correct at least one validation error of the one or more validation errors; xii) store the one or more validation errors for further review; xiii) receive a first plurality of rules from the first destination system; xiv) receive a second plurality of rules from the second destination system; xv) compare the first plurality of rules to the second plurality of rules; xvi) generate the plurality of rules based on the comparison; xvii) detect a first conflict between the first plurality of rules and the second plurality of rules associated with a first data field; xviii) generate a format for the data message to include a first version of a first data field for the first destination system and a second version of the first data field for the second destination system; xix) retrieve a data value from the source database; xx) convert the data value from a first format to a second format; xxi) populate a data field of the data message with the converted data value; xxii) retrieve a data value from the source database; xxiii) populate a first data field of the data message with the data value, wherein the first data field is associated with a first destination system of the plurality of destination systems; xxiv) convert the data value from a first format to a second format; and xxv) populate a second data field of the data message with the converted data value, wherein the second data field is associated with a second destination system of the plurality of destination systems.

[0006] At least one advantage of this system is that because the data is validated prior to being transmitted to and ingested by destination systems. This system validates requirements from producer and consumers concurrently. This provides transparent data validation requirements to all data producers and consumers by performing point-in-time evaluation for data quality, content, and condition. The system also provides an externalized validation system to the calling application which allows configurability on invocation. This system also provides a dashboard view and error logging provides holistic view of data quality.

[0007] In one aspect, a computer system for performing enhanced rules conflict checking may be provided. The computer system includes at least one processor in communication with at least one memory device. The at least one processor is programmed to store a plurality of rules for transmitting to a plurality of destination systems and receive, from a requesting system, a data message for transmission to the plurality of destination systems. The data message is configured to be routed to each of the plurality of destination systems. The at least one processor is also programmed to compare the data message to the plurality of rules to validate the data message and if the data message is validated, instruct the requesting system to transmit the data message to the plurality of destination systems. The computer system may have additional, less, or alternate functionality, including that discussed elsewhere herein.

[0008] In another aspect, a computer-based method for performing enhanced rules conflict checking may be provided. The method may be implemented a computer system including at least one processor in communication with at least one memory device. The method includes storing, in the at least one memory device, a plurality of rules for transmitting to a plurality of destination systems and receiving, from a requesting system, a data message for transmission to the plurality of destination systems. The data message is configured to be routed to each of the plurality of destination systems. The method also includes comparing the data message to the plurality of rules to validate the data message and if the data message is validated, instructing the requesting system

to transmit the data message to the plurality of destination systems. The method may include additional, less, or alternate actions, including those discussed elsewhere herein.

[0009] In a further aspect, a computer system for performing over-time event monitoring is provided. The computer system includes at least one processor in communication with at least one memory device. The at least one processor is programmed to scan a plurality of data messages. The at least one processor is also programmed to detect a first data message in the plurality of data messages. The first data message is associated with a first event that includes the first data message and a second data message. The at least one processor is further programmed to scan the plurality of data messages for the second data message associated with the first event. If the second data message is not detected based on the scan, the at least one processor is also programmed to determine whether the first event has expired. If the first event has expired, the at least one processor is further programmed to transmit an error message to a user computer device. The computer system may have additional, less, or alternate functionality, including that discussed elsewhere herein.

[0010] In a still further aspect, a method for performing over-time event monitoring is provided. The method is performed by a computer system including at least one processor in communication with at least one memory device. The method includes scanning a plurality of data messages. The method also includes detecting a first data message in the plurality of data messages. The first data message is associated with a first event that includes the first data message and a second data message. The method further includes scanning the plurality of data messages for the second data message associated with the first event. If the second data message is not detected based on the scan, the method also includes determining whether the first event has expired. If the first event has expired, the method further includes transmitting an error message to a user computer device. The method may include additional, less, or alternate actions, including those discussed elsewhere herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The Figures described below depict various aspects of the systems and methods disclosed therein. It should be understood that each Figure depicts an embodiment of a particular aspect of the disclosed systems and methods, and that each of the Figures is intended to accord with a possible embodiment thereof. Further, wherever possible, the following description refers to the reference numerals included in the following Figures, in which features depicted in multiple Figures are designated with consistent reference numerals.

[0012] There are shown in the drawings arrangements which are presently discussed, it being understood, however, that the present embodiments are not limited to the precise arrangements and are instrumentalities shown, wherein:

[0013] FIG. 1 illustrates an exemplary system for data validation and integrity checks in accordance with at least one embodiment.

[0014] FIG. 2 illustrates another exemplary system for data validation and integrity checks in accordance with at least one embodiment.

[0015] FIG. 3 illustrates an exemplary system for compiling rules for making data validation and integrity checks using the system shown in FIG. 1.

[0016] FIG. 4 illustrates another exemplary system for compiling rules for making data validation and integrity checks using the system shown in FIG. 2.

[0017] FIG. 5 illustrates a process for performing data validation and integrity checks using at least one of the system shown in FIG. 1, the system shown in FIG. 2, and the system shown in FIG. 4.

[0018] FIG. 6 illustrates a process for compiling rules for making data validation and integrity

checks using the system shown in FIG. 3.

[0019] FIG. 7 illustrates an exemplary system for monitoring events that consist of a plurality of data messages in accordance with at least one embodiment.

[0020] FIG. 8 illustrates a process for monitoring events that consist of a plurality of messages the system shown in FIG. 7.

[0021] FIG. 9 illustrates an example configuration of user computer device used in the system shown in FIG. 1, in accordance with one example of the present disclosure.

[0022] FIG. 10 illustrates an example configuration of a server computer device used in the system shown in FIG. 1, in accordance with one example of the present disclosure.

[0023] The Figures depict preferred embodiments for purposes of illustration only. One skilled in the art will readily recognize from the following discussion that alternative embodiments of the systems and methods illustrated herein may be employed without departing from the principles of the invention described herein.

DETAILED DESCRIPTION OF THE INVENTION

[0024] The present embodiments may relate to, inter alia, systems and methods for providing over-time event monitoring with enhanced rules conflict check with data validation. In an exemplary embodiment, the process is performed by a data integrity and validation (“DIV”) computer device, also known as a data integrity and validation (“DIV”) server, and/or an event message analysis (“EMA”) computer device, also known as an event message analysis (“EMA”) server.

[0025] The DIV computer device interacts with a database or data structure that provides data to a data flow pipeline. The DIV computer device is programmed to provide data to one or more destination systems. The DIV computer device is also programmed to identify data anomalies to allow the user to take corrective measures before the data anomalies are introduced to the destination systems or to notify the user that non-critical data anomalies. Data anomalies include where the data in a data message conflicts with the rules for that data message. Data anomalies can include, but are not limited to, missing data, incorrect data, improperly formatted data, data in the wrong place, and other errors that could impact other systems.

[0026] The DIV computer device is configured to evaluate the valuation requirements from the data producers and the data consumers (i.e., the destination systems) to determine if the data is in the proper format and location in data messages to be transmitted to the destination systems. By consolidating the data validation and data consumption rules for a plurality of destination computer systems, the DIV computer device is able to concurrently execute data validation for multiple systems, provide point-in-time evaluation for data quality, content, and condition, provide analysis of data structures across time periods, provide data validation externalized from the calling application or system, and provide dashboard and error logging analysis for an overall view of the quality of the data being provided.

[0027] The DIV system is also programmed to assist with determining the format of a data message based on the rules and requirements of the destination systems that the data message is to be routed to. The DIV system receives the rules and requirements from each of the destination systems that a data message is to be routed to. The DIV system compares the rules and requirements to each other to determine the final rules and requirements for the data message. The DIV system also determines the layout of the payload of the data message so that all of the required fields and the required formats are included in the data message. The DIV system instructs the destination systems in which fields each destination system's data fields are stored. When the destination system receives the data message, the destination system knows which fields to extract the data from.

[0028] The DIV system can also be in communication with an event message analysis (EMA) system. The EMA system stores the data messages that are transmitted from the source system to the destination systems. Some of the messages are associated with one event or another. However, these events may be mostly automated and require multiple messages to complete. Furthermore, the messages may take place separately over a period of time, such as a few hours, a few days, a

few weeks, or any other period of time. For example, one event could be the renewal of a policy or subscription. This event could require four separate messages to be transmitted to the destination systems over a period of 14 days. If all four of the messages are not transmitted within fourteen days of the first message being transmitted, the event fails.

[0029] In the exemplary embodiment, the EMA system stores all of the messages in a database. In other embodiments, the EMA system only stores those messages associated with specific events, such as those events the EMA system is analyzing. In some embodiments, the EMA system deletes stored messages after a predetermined period of time, such as six months. The EMA system performs an analysis on the plurality of stored messages. The EMA system searches the plurality of stored messages for event starting messages, these messages indicate the start of an event. When the EMA system detects an event starting message, the EMA system scans the rest of the plurality of stored messages for the other messages in the event. If all of the messages associated with the event are present, then the event is marked completed. If one or more messages associated with the event are not present, then the event is marked incomplete. The EMA system then checks the deadline or expiration date for the event. If the event is incomplete and past the deadline, the EMA system issues a failure message. If the event is incomplete and close to, but not past, the deadline, the EMA system may send a warning message to indicate the incomplete event.

[0030] In some embodiments, the EMA system analyzes the plurality of stored messages on a periodic basis, such as once a day or once an hour. In other embodiments, the EMA system provides analysis on an ongoing basis. In some embodiments, the EMA system tracks a plurality of events and updates the event tracking as new messages are analyzed. In other embodiments, the EMA system performs an analysis on all of the plurality of stored messages every time the analysis is performed.

[0031] FIG. 1 illustrates an exemplary system **100** for data validation and integrity checks in accordance with at least one embodiment. In system **100**, a source system **105** is in communication with an information source database **110**. The source system **105** is also in communication with a data integrity and validation (DIV) system **115**, which is associated with a rules database **120**. The source system **105** is further in communication with one or more destination systems **125-135**.

[0032] The destination systems A-C **125-135** represent multiple systems that request information from the information source database **110**. In system **100**, the three destination systems A-C **125-135** are in communication in a serial order, where a data message is transmitted from the source system **105** to destination system A **125**. In some embodiments, destination systems A-C **125-135** are independent of each other. In other embodiments, destination systems A-C **125-135** are related. For the discussion here, destination system C **135** is considered to be ‘downstream’ of destination system B **130**, which is ‘downstream’ of destination system A **125**. While only three destination systems **125-135** are shown in the Figures presented herein, the systems and methods described herein may be used for any number of destination systems **125-135**.

[0033] The data message is then transmitted from destination system A **125** to destination system B **130** and then from destination system B **130** to destination system C. When the data message reaches each destination system **125-135**, the corresponding destination system **125-135** retrieves the data that it requires from the data message and ignores the data in the rest of the fields. The data message needs to include the required information for each of the destination systems **125-135** that the data message will be routed to. For example, if the data message is being routed to destination system A **125**, then destination system B **130**, and finally destination system C **135**, the data message will include all of the data required by each destination system **125-135**. This also requires the data message to conform to the rules and requirements for each of the destination systems **125-135**.

[0034] Some of the data fields may have contradictory requirements for different destination systems **125-135**. For example, destination system A **125** may require the date to be in the American format of month, day, and year, but destination system B **130** may require the data to be

in the European format of day, month, and year. Or destination system B **130** may require the year to be two digits long, while destination system C **125** may require the year to be four digits long. Or the state could be required to be spelled out instead of using the two letter code. Or the zip code may require five digits or nine digits. The same data in a data field could be required to be 8 bits long for destination system A **125** and 12 bits long for destination system B **130**. In some embodiments, this is resolved by having two different data fields in the data message, one field including the information for destination system A **125** and the other field including the information for destination system B **130**.

[0035] In the exemplary embodiment, the source system **105** is a data source computer device that receives requests for data and generates data messages to respond with. The source system **105** transmits the requested data to one or more of the destination systems **125-135** in the appropriate data message. The source system **100** can transmit a data message in response to a received data request from one or more destination systems **125-135**. The source system **100** can also transmit a data message at a predetermined time or after a predetermined period of time. The source system can further receive a request to transmit the data to the destination systems **125-135** from an outside system, such as when specific data is received in the information source database **110**.

[0036] The data message can include, but is not limited to, a header field, a message type field, one or more destination fields, a length field, and a payload. The payload includes a plurality of data fields populated from the information source database **110**. As described further below, the length field and the data fields in the payload of each data message depend on the message type and the destination systems **125-135** that the data message is being transmitted to.

[0037] The information source database **110** is configured to store information, such as, but not limited to, client records, property records, driving records, financial records, insurance policy information, cybersecurity information, and/or any other type of data required. The source system **105** is programmed to provide the information in the information source database **110** to the destination systems **125-135**. In some embodiments, the records in the information source database **110** are formatted based on how the information is to be presented to the destination systems **125-135**. For example, the records in the information source database **110** are formatted so that the information can be directly added to the message providing the information to the destination systems **125-135**. In one example, a date is formatted in the European style, where the number for the date is before the month. In other embodiments, the source system **105** converts or formats the information prior to adding the information to the data message to the one or more of destination systems **125-135** based on the requirements of the receiving destination systems **125-135**. The data in the information source database **110** can be provided from a plurality of different computer systems. In some embodiments, the data is provided to the source system **105**, which stores the data in the information source database **110**. In other embodiments, other computer systems store the data in the information source database **110**. While the other computer systems may validate the information as it is being provided, the DIV system **115**, described below, also validates the data in the information source database **110** to ensure that the data is placed into the data message correctly.

[0038] The DIV system **115** is programmed to receive the data message for validation before the data message is transmitted to the destination systems **125-135**. The DIV system **115** is in communication with rules database **120**. The rules database **120** includes the rules associated with the destination systems **125-135** about how those destination systems **125-135** expect to receive the data stored in the information source database **110**. The DIV system **115** compares the data message to the rules in the rules database **120** to validate the data message. If the data message is validated, then the DIV system **115** notifies the source system **105** of the validation. The source system **105** can then transmit the data message to the destination systems **125-135**. If the data message is not validated, then the DIV system **115** determines why the data message was not validated. If the DIV system **115** determines that the validation issue is minor, the DIV system **115**

can log the validation issue and instruct the source system **105** to transmit the data message to the destination systems **125-135**. If the DIV system determines that the validation issue is significant, then the DIV system **115** can log the validation issue and then notify the user of the validation issue. In some embodiments, the user can then correct the validation issue to allow the data message to be transmitted. In some further embodiments, the DIV system **115** corrects the validation issue in the data message and transmits the corrected data message to the source system **105** to be transmitted to the destination systems **125-135**.

[0039] In some embodiments, a minor (or low) validation error may include data that is not vital to the operation of system **100**. A minor validation error may also be considered a low risk error, where the validation failure presents an acceptable risk, but the error should be further investigated and repaired at a later time. For example, data for trend analysis may be missing. As the trend analysis only occurs on a periodic basis, the validation error can be considered minor as the rest of the system **100** will continue to operate effectively. Furthermore, since the error is logged, those performing trend analysis could be aware of the issue before they begin performance of the analysis and thus have time to correct the issue. An example of a significant (or high level) validation error may include missing data that is vital to the operation of the system **100**, such as, but not limited to, financial data. A significant validation error may also be considered a high risk error, where the validation failure presents an unacceptable risk and could critically impact the destination systems **125-135**, where data correction is required before the data message is released. If this data is missing or incorrect, then one or more of the downstream systems **125-135** may operate inefficiently or may introduce errors into the system **100**. Accordingly, significant or high level validation errors are logged and the associated data message is not transmitted. In some embodiments, the low risk issues could be considered at a warning level, while the high risk issues are at an error level.

[0040] In validating the data message, the DIV system **115** can evaluate a plurality of rules to ensure that each of the plurality of rules is being followed for the data message, so that the data message will not cause any problems with the destination systems **125-135**. Example rules can include, but are not limited to, i) quality rules (i.e. is the data present? does the data element meet the expected requirements for format and length?); ii) content rules (i.e. allowed values); iii) conditional rules (i.e. evaluation of secondary elements for quality and/or content based on the presence or content of primary elements); and/or any other types of rules.

[0041] Examples of quality rules include, but are not limited to, ensuring that the birthday field is filled, ensuring that the date is formatted correctly (i.e. two digits in each of the month, day, and year fields), and ensuring that the proper field type is used (i.e. Boolean, text, integer, etc.). Examples of content rules include, but are not limited to, ensuring that the provided date is proper (i.e. no greater than 28 days in February 1990). Examples of conditional rules include, but are not limited to (i.e. if field A is filled, then field B needs to be filled; if field A is set to 8, then field B needs to be filled; or if a policy states that it is for two people, but there is only information for one person). Other rules may be used based on the requirements of different destination systems **125-135**.

[0042] In some embodiments, DIV system **115** has access to the information source database **110** to check if the correct information was placed into the fields of the data message. For example, the DIV system **115** can compare the fields of the data message to the raw data in the information source database **110** and determine that the source system **105** improperly converted the data prior to placing the data in the data message. In some further embodiments, the DIV system **115** logs the improper conversion issue to be repaired. In some embodiments, the destination system **125-135** provides preferences on how to handle issues and errors. For example, the destination system **125-135** may instruct the source system **105** to transmit the data message even if there are specific types of errors or errors in specific fields of the data message. In still further embodiments, the DIV system **115** repairs the improperly converted field so that the data message can still be transmitted

to the destination systems **125-135**.

[0043] The rules database **120** stores a plurality of rules based on the rules of each of destination systems **125-135**. The rules in the rules database **120** describe many or all of the fields in the data message including, but not limited to, what information should be included in each field, when, and in what format.

[0044] In some embodiments, the DIV system **115** transmits a validation success or failure notification to the source system **105** after analyzing the data message. In other embodiments, the DIV system **115** transmits the data message to the destination system(s) **125-135**, after analyzing the data message. The validation notification can include information about all of the low risk and high risk issues that were detected by the DIV system **115**.

[0045] FIG. 2 illustrates another exemplary system **200** for data validation and integrity checks in accordance with at least one embodiment. In system **200**, destination systems A **125** and B **130** are in parallel with each other, while destination system C **135** is downstream of destination system B **130**. When a data message is transmitted to destination system B **130**, the data message must comply with the rules of destination system B **130** and destination system C **135**. In some embodiments, the same data message is transmitted to destination system A **125** and destination system B **130** simultaneously and must comply with the rules for both. In other embodiments, different data messages are sent to destination system A **125** and destination system B **130**. Thus the different data messages only have to comply with the rules of any destination systems **125-135** that the data message is routed to.

[0046] In other embodiments, other configurations of destination systems **125-135** may be used with the systems and methods described herein. Furthermore, in at least one embodiment, destination system A **125** may provide information for destination system C **135**. For example, a data field in the payload may be empty when the data message is transmitted from source system **105** to destination system A **125**. Destination system A **125** can then fill in the appropriate data into the data field of the data message, so that destination system C **135** can receive that data. In this example, destination system A **125** may be a pricing system and determines and provides a price for destination system C **135** to use.

[0047] FIG. 3 illustrates an exemplary system **300** for compiling rules for making data validation and integrity checks using the system **100** (shown in FIG. 1).

[0048] In system **300**, the DIV system **115** is in communication with a plurality of destination systems **125-135**. The DIV system **115** receives a plurality of rules from each of the destination systems **125-135**. The DIV system **115** analyzes the plurality of rules to determine if there are any rules conflicts. Conflicts can include, but are not limited to, rules requiring data to be in different formats for different destination systems **125-135**, missing data, different versions of the effectively the same data (birthdate vs. current age), and contradictory data.

[0049] The DIV system **115** is also in communication with the information source database **110**. In some embodiments, the DIV system **115** compares the plurality of rules to the data stored in the information source database **110**. The DIV system **115** also determines if there are any rules conflicts between the plurality of rules and the information stored in the information source database **110**. For example, a date may be stored in European format vs. American format; the birthdate may be stored, but not the current age; or the number of digits required for the year may vary. For both of these examples, the DIV system **115** can instruct the source system **105** to properly format the required data as the source system **105** generates the data message.

[0050] The DIV system **115** compares the plurality of rules from each of the destination systems **125-135** and the available data in the information source database **110** and generates a blueprint for the data message. The DIV system **115** determines which fields and field types will be put in the payload section, where those fields will be located in the data message, and how those fields will be formatted. The DIV system **115** also informs the destination systems **125-135** where the data that each of the destination systems **125-135** needs will be located in the data message.

[0051] For example, the three fields for destination system A **125** are fields 1, 2, and 3. The three fields for destination system B **130** are fields 1, 2, and 4. And the four fields for destination system C **135** are fields 1, 2, 3, and 5. The DIV system **115** ensures that the data message includes all of the required fields in the required formats. In this example, field 3 could be the date of activation of a policy in the American date format. Destination systems A **125** and C **135** require the date field in the American date format. However, destination system B **130** requires the date field in the European date format. Since destination system B **130** receives the same data message as destination systems A **125** and C **135**, the DIV system **115** generates the data message to store the date in the American format in field 3 and in the European format in field 4. While this adds an additional field to the data message, it reduces the number of messages required to be generated and sent over the network, thus reducing overall network traffic.

[0052] In some other embodiments, the fields of the data message are predetermined, where the DIV system **115** does not have access to the information source database **110**. In these embodiments, the data message is designed to fit the requirements of each of the destination systems **125-135**. Then when another destination system is added at a subsequent point in time, such as destination system D (not shown), the DIV system **115** receives the requirements of destination system D and compares those requirements to the data message. If the requirements are compatible, then no other actions are required and destination system D is integrated into the system **100**. If there are issues or conflicts, then the DIV system **115** can notify one or more administrator users to fix the conflicts. In further embodiments, the DIV system **115** can edit the data message and inform destination system D where the fields that it needs to access are located in the data message.

[0053] FIG. 4 illustrates a further system **400** for data validation and integrity checks in accordance with at least one embodiment. In system **400**, source system **105** has access to two information source databases **110**, information source database A **405** and information source database B **410**. In some embodiments, each database **405-410** provides different information and is associated with different data messages.

[0054] In system **400**, the source system **105** routes the appropriate information from the appropriate information source database **405-410** to the appropriate destination system(s) **125-135**. For example, source system **105** is responsible for four separate data messages. The first data message includes data from information source database A **405** and is transmitted to destination systems A **125** and B **130**. The DIV system **115** receives the rules requirements from destination systems A **125** and B **130** for the first data message. The DIV system **115** generates the first data message in response to the received rules sets and the information available in information source database A **405**. When source system **105** receives a request for the first data message, the source system **105** populates the first data message with the appropriate data from information source A **405**. Then source system **105** requests that the DIV system **115** validate the populated first data message. The DIV system **115** uses the rules sets from destination system A **125** and destination system B **130** to validate the first data message. If the first data message is validated, the source system **105** transmits the first data message to destination system A **125**.

[0055] The second data message includes data from information source database A **405** and is transmitted to destination systems A **125**, B **130**, and C **135**. The DIV system **115** receives the rules requirements from destination systems A **125**, B **130**, and C **135** for the first data message. The DIV system **115** generates the second data message in response to the received rules sets and the information available in information source database A **405**. The second data message can include more fields than the first data message, which may include additional data requested by destination system C **135** and/or additional fields needed to resolve conflicts between the rules for destination systems A **125** or B **130** and destination system C **135**.

[0056] When source system **105** receives a request for the second data message, the source system **105** populates the second data message with the appropriate data from information source A **405**.

Then source system **105** requests that the DIV system **115** validate the populated second data message. The DIV system **115** uses the rules sets from destination system A **125**, B **130**, and C **135** to validate the second data message. If the second data message is validated, the source system **105** transmits the second data message to destination system A **125**.

[0057] The third data message includes data from information source database B **410** and is transmitted to destination systems A **125** and C **135**. The DIV system **115** receives the rules requirements from destination systems A **125** and C **135** for the third data message. The DIV system **115** generates the third data message in response to the received rules sets and the information available in information source database B **410**. When source system **105** receives a request for the third data message, the source system **105** populates the third data message with the appropriate data from information source B **410**. Then source system **105** requests that the DIV system **115** validate the populated third data message. The DIV system **115** uses the rules sets from destination system A **125** and destination system C **135** to validate the third data message. If the third data message is validated, the source system **105** transmits the third data message to destination system A **125**.

[0058] The fourth data message includes data from information source database B **410** and is transmitted to destination systems B **130** and C **135**. The DIV system **115** receives the rules requirements from destination systems B **130** and C **135** for the third data message. The DIV system **115** generates the third data message in response to the received rules sets and the information available in information source database B **410**. When source system **105** receives a request for the fourth data message, the source system **105** populates the fourth data message with the appropriate data from information source B **410**. Then source system **105** requests that the DIV system **115** validate the populated fourth data message. The DIV system **115** uses the rules sets from destination system B **130** and destination system C **135** to validate the fourth data message. If the fourth data message is validated, the source system **105** transmits the fourth data message to destination system B **130**.

[0059] FIG. 5 illustrates a process **500** for performing data validation and integrity checks using at least one of the system **100** (shown in FIG. 1), the system **200** (shown in FIG. 2), and the system **400** (shown in FIG. 4). In the exemplary embodiment, process **500** is performed by the source system **105** and the DIV system **115** (both shown in FIG. 1).

[0060] The source system **105** receives **505** a data request to be transmitted to one or more destination system **125-135** (shown in FIG. 1). In some embodiments, the data request could originate from one or more of the destination systems **125-135**. In other embodiments, the data request could be automated, where the data request or the data message is transmitted on a regular basis. In other embodiments, the data request could originate from a user computer device, such as a user computer device being used by a user to input information.

[0061] The source system **105** determines which data message to respond with. The data message selected may be based on one or more of the destination systems **125-135** receiving the data message, the information source database **110** (shown in FIG. 1) containing the data, and/or the data request message.

[0062] The source system **105** determines **510** one or more destination systems **125-135** to transmit the data message from the data request. As described above, the source system **105** can store a plurality of data messages that can be routed to different destination systems **125-135**. In at least one embodiment, each data message is routed to a specific set of destination systems **125-135** due to the required fields from the requirements of the associated destination systems **125-135**. By determining **510** the destination systems, the source system **105** also can determine the appropriate data message to transmit to those destination systems **125-135**.

[0063] The source system **105** builds **515** the data message based on the data request. In the exemplary embodiment, the format and fields of the data message have been previously determined, such as in process **600** (shown in FIG. 6). The source system **105** populates the fields

of the data message with the data from the information source database **110**. In some embodiments, the source system **105** converts some of the data from the information source database **110** to a required format prior to populating that data in the appropriate data field. For example, if the field requires that the year be listed in two digits and the information source database **110** stores the year using four digits, the source system **105** converts the year field from four to two digits for the data message. Or if the data field requires the age of the client, but only the birthdate is stored in the information source database **110**, the source system **105** determines the age of the client from the birthdate and the current date.

[0064] The DIV system **115** compares **520** the data message to a plurality of rules associated with the determined one or more destination systems **125-135**. In some embodiments, the source system **105** calls the DIV system **115** to validate the data message. In other embodiments, the source system **105** transmits the data message to the DIV system **115**.

[0065] The DIV system **115** validates **525** the data message based on the plurality of rules stored in the rules database **120** (shown in FIG. 1). In some embodiments, the rules database **120** stores the rules for each destination system **125-135**. In other embodiments, the rules database **120** stores the rules for each data message, where the rules for each destination system **125-135** that the data message is scheduled to go to has been integrated together and stored in the rules database **120**, such as in process **600**.

[0066] If the data message is validated, the data message is transmitted **530** to the one or more destination systems **125-135**. In some embodiments, the source system **105** transmits **530** the data message. In other embodiments, the DIV system **115** transmits **530** the data message.

[0067] If the data message is not validated, the DIV system **115** determines **535** the level of error associated with the validation failure. If the level of error is low (or minor), the DIV system **115** logs **540** and the data message is transmitted **530** to the one or more destination systems **125-135**. If the error level is high (or significant), the DIV system **115** logs **545** the error and requests **550** user input. While the error system described herein as high level and low level errors, in some embodiments, other error valuation systems, such as colors or a scale may be used with the systems described herein.

[0068] In some embodiments, the user is in the process of inputting data, such as through the source system **105**. The user may be entering data to request a policy or policy update. The DIV system **115** determines that one or more fields in the data message are missing or are incorrectly formatted. In these embodiments, the DIV system **115** may inform the source system **105** of the error and the source system **105** may request corrected information from the user. The user input may correct the error, so that the data message may then be transmitted **530**.

[0069] In some other embodiments, the user is an administrator makes one or more changes to at least one of the data message and the information source database **110** for future data messages to correct the error.

[0070] In the exemplary embodiment, the DIV system **115** compares **520** the entire data message to the rules associated with that data message. Then the DIV system **115** compiles all of the issues and errors for the data message.

[0071] In some embodiments, the destination system **125-135** provides preferences on how to handle issues and errors. For example, the destination system **125-135** may instruct the source system **105** to transmit **530** the data message even if there are specific types of errors or errors in specific fields of the data message. In other embodiments, the source system **105** transmits information about the issues and errors to the destination system(s) **125-135** associated with the error or issue and request instructions on how to proceed. The corresponding destination system(s) **125-135** instructs the DIV system **115** and/or the source system **105** whether to transmit **530** the data message, fix the issue or error, or prevent transmission of the data message.

[0072] If there are multiple errors, then each error is logged **540-545**. If all of the errors are low, then the data message is transmitted **530**.

[0073] In some embodiments, the DIV system **115** corrects the error.

[0074] FIG. **6** illustrates a process **600** for compiling rules for making data validation and integrity checks using the system **300** (shown in FIG. **3**). In the exemplary embodiment, process **600** is performed by the DIV system **115** (shown in FIG. **1**). Process **600** can be used when a new data message is being designed. For the new data message the destination systems **125-135** that the message will be transmitted to are analyzed for their rules to determine if there any conflicts in the fields of the data message. The rules describe the data requirements of the destination systems **125-135** that the data message will be transmitted to, including i) quality rules (i.e. is the data present? does the data element meet the expected requirements for format and length?); ii) content rules (i.e. allowed values); iii) conditional rules (i.e. evaluation of secondary elements for quality and/or content based on the presence or content of primary elements); and/or any other types of rules necessary. The data message is designed so that each destination system **125-135** will know where in the data message its fields are stored.

[0075] The DIV system **115** receives **605** a first set of rules from a first destination system, such as destination system A **125** (shown in FIG. **1**). The DIV system **115** receives **610** a second set of rules from a second destination system, such as destination system B **130**. While only two sets of rules are described in process **600**, multiple sets of rules may be used to create the data message, these can include, but are not limited to, rules from the destination systems **125-135**, rules from the information source database **110** (shown in FIG. **1**), and rules for the data message itself.

[0076] The DIV system **115** compares **615** the first set of rules and the second set of rules. In some embodiments, the DIV system **115** compares **615** the rules to each other. In other embodiments, the DIV system **115** also compares **615** the rules to the available data in the information source database **110**. The DIV system **115** compares the rules to detect **620** rules conflicts. The rules conflicts can include, but are not limited to, quality rules conflicts (where the required data is missing or misformatted); content rules conflicts (where the data available is outside of the required bounds); mismatching rules conflicts (where different sets of rules require data to be in different formats); and differing data requirements (where different data is required by different destination systems **125-135**). When the DIV system **115** detects **620** a rules conflict, the DIV system **115** determines **625** a potential fix. For example, the DIV system **115** may determine **625** to create two different fields with the same information but in different formats. The DIV system **115** may also determine **625** that a specific piece of data is not provide, but can be derived from the available information. The DIV system **115** stores **630** the potential fix for the rules conflict and returns to Step **615** to compare **615** the sets of rules. The DIV system **115** continues to cycle through Steps **615-630** until all of the rules conflicts are resolved. In some embodiments, the DIV system **115** is unable to determine **625** a potential fix for a rules conflict. In these embodiments, the DIV system **115** logs the rules conflict and requests user input to correct the rules conflict. For example, a required field or piece of information might not be available in the information source database **110**. In this example, the user can remove the requirement, ensure that the required data is added to the information source database **110**, or instruct that the data is retrieved from a different source than information source database **110**.

[0077] Once the DIV system **115** detects **620** no more unresolved rules conflicts, the DIV system **115** generates **635** the format for the data message based on the comparison and any potential fixes. In some embodiments, the DIV system **115** generates **635** the format of the data message and the instructions on how to populate the fields of the data message from the information source database **110**. The source system **105** would then follow those instructions in populating the data message as shown in Step **515** (shown in FIG. **5**). Instructions can include, but are not limited to, which fields to place where in the data message, how to format those fields, and how to generate the data for the fields using available data from the information source database **110**.

[0078] In some embodiments, the DIV system **115** instructs the information source database **110** to store some information differently, such as in a different format, to allow the source system **105** to

quickly and efficiently build **515** the data message.

[0079] In some embodiments, the DIV system **115** transmits the data locations of the data fields in the data message to each of the downstream systems **125-135**, so that each one knows which fields to extract data from. In some embodiments, the destination systems **125-135** receive a complete mapping of the data message. In other embodiments, the destination systems **125-135** receive the mappings of the data fields that are to be retrieved by the corresponding destination system **125-135**.

[0080] FIG. 7 illustrates an exemplary system **700** for monitoring events that consist of a plurality of data messages in accordance with at least one embodiment. In the exemplary embodiment, system **700** is an enhancement or add-on to one or more of system **100** (shown in FIG. 1), system **200** (shown in FIG. 2), and system **400** (shown in FIG. 4).

[0081] Events are groups of associated messages that are transmitted to the destination systems **125-135**. For example, an event for renewing a policy or subscription may require four messages to complete. These four messages may include different information, be provided from different systems **100**, and be transmitted to different destination systems **125-135**.

[0082] In system **700**, one of source system **105** and DIV system **115** transmit data messages that are being transmitted to destination systems **125-135** to be stored in an event message analysis (EMA) database **710**. As the data message is being transmitted to the destination systems **125-135**, the data message is cloned or copied and transmitted to the EMA database **710** for storage. In some embodiments, the cloned data message is transmitted to the EMA system **705** to determine whether or not to store the data message. In some embodiments, the EMA system **705** acts as a filter and only stores those messages that are associated with events. In these embodiments, the EMA system **705** can determine whether a message is associated with an event by one or more of, the message type, the destination systems **125-135** that the message is being routed to, the source system **105** that is transmitting the data message, and/or one or more payload fields of the data message.

[0083] In some embodiments, the EMA database **710** only stores data messages that have been validated by the DIV system **115**. In other embodiments, the EMA database **710** stores all of the data messages that source system **105** generates. When a data message is received, the EMA database **710** stores that data message for future analysis. The EMA system **705** analyzes the plurality of data messages stored in the EMA database **710** to determine if all of the messages associated with an event have been transmitted in the required amount of time. For this example, after the first data message for the policy or subscription renewal is transmitted, the systems **100** have 14 days to complete the renewal process and transmit the remaining three messages.

[0084] The EMA system **705** detects the first data message after it is transmitted to the destination systems **125-135**. In some embodiments, the EMA system **705** analyzes the data messages in the EMA database **710** on a periodic basis, such as a daily or hourly basis. In some embodiments, the EMA system **705** analyzes all of the data messages since the last analysis. In other embodiments, the EMA system **705** analyzes all of the data messages in the EMA database **710**. In further embodiments, the EMA **705** analyzes the data messages as they are being transmitted to the destination systems **125-135**.

[0085] The EMA system **705** detects the first data message of the event and then scans the rest of the data messages for the other three data messages associated with the event. In some embodiments, the EMA system **705** tracks a list of open or active events, where the EMA system **705** scans the plurality of data messages for the data messages associated with each open or active event. The EMA system **705** can store indicators for each event that notes when a message associated with an event is transmitted. In the above example, the EMA system **705** can store that the first, second, and third messages associated with the renewal event have been transmitted, but not the fourth message. The EMA system **705** can also store the dates and times associated with each message for the event. In some embodiments, the event needs to have all four messages transmitted to the destination systems **125-135** to complete the event within a specific period of

time after the first message. In other embodiments, the four messages must be transmitted in order for the event to be properly executed. In these embodiments, the EMA system **705** can determine when a message is missing or is out of order. The EMA system **705** can also determine if one message was transmitted within a specific time period after another message. For example, the second message may need to arrive within two hours of the first message and the third message needs to arrive within four hours of the second message.

[0086] When the EMA system **705** detects an event starting message, the EMA system **705** scans the rest of the plurality of stored messages for the other data messages in the event. If all of the data messages associated with the event are present, then the event is marked completed. If one or more data messages associated with the event are not present, then the event is marked incomplete. The EMA system **705** also checks the deadline or expiration date for the event. If the event is incomplete and past the deadline, the EMA system **705** issues a failure message, such as to a user computer device. If the event is incomplete and close to, but not past, the deadline, the EMA system **705** can send a warning message to indicate the incomplete event to the user computer device.

[0087] In some embodiments, the EMA system **705** is associated with the DIV system **115**. In other embodiments, the EMA system **705** is a part of the DIV system **115**.

[0088] In the exemplary embodiment, the EMA system **705** and database **710** are in communication with a plurality of systems **100** and corresponding source systems **105**. In this way, system **700** receives data messages from a plurality of different systems **100** to allow the EMA system **705** to analyze all or a significant portion of the messages that are being transmitted to the destination systems **125-135**. In other embodiments, the EMA system **705** receives the data messages from one or more of the destination systems **125-135**.

[0089] In some embodiments, the rules for events could be stored in the EMA database **710**. In other embodiments the rules for events and associated messages could be stored in a rules database, such as rules database **120**.

[0090] FIG. **8** illustrates a process **800** for monitoring events that consist of a plurality of messages the system **700** (shown in FIG. **7**). In the exemplary embodiment, process **800** is performed by the EMA system **705** (shown in FIG. **7**). In some embodiments, process **800** is performed by the DIV system **115** (shown in FIG. **1**) and/or the EMA system **705**.

[0091] In the exemplary embodiment, the EMA system **705** scans **805** a plurality of data messages. The plurality of data messages can be stored in the EMA database **710**. In the exemplary embodiment, the plurality of data messages include data messages that were previously validated by the DIV system **115** and then transmitted to the plurality of destination systems **125-135**.

[0092] In the exemplary embodiment, the EMA system **705** detects **810** a first data message in the plurality of data messages. The first data message is associated with a first event that includes the first data message and a second data message. The first event can include a plurality of events. For the purposes of this illustration the second data message represents the last data message of the event.

[0093] In the exemplary embodiment, the EMA system **705** scans **815** the plurality of data messages for the second data message associated with the first event.

[0094] If the second data message is not detected based on the scan, the EMA system **705** determines **820** whether the first event has expired.

[0095] If the first event has expired, the EMA system **705** transmits **825** an error message to a user computer device.

[0096] If the second data message was detected based on the scan, the EMA system **705** indicates that the first event is complete. In some embodiments, the EMA system **705** tracks a current status of a plurality of events based on a plurality of first messages. The EMA system **705** tracks each data message that is transmitted based on the associated event. When all of the data messages associated with an event have been transmitted and detected by the EMA system **705**, the EMA

system **705** indicates that the event is complete. In some embodiments, the EMA system **705** transmits a message to a user computer device indicating the successful completion of the first event. In other embodiments, the EMA system **705** stores an indicator that the first event was successfully completed. In still further embodiments, the EMA system **705** deletes the data messages associated with the first event and the indicators of the first event, when the first event is determined to be complete.

[0097] If the second data message was detected based on the scan, the EMA system **705** determines if there are one or more additional data messages associated with the first event. For example if an event has four data messages associated with that event that need to be transmitted for the event to be complete. If there are one or more additional data messages associated with the first event, the EMA system **705** scans the plurality of data messages to detect the one or more additional data messages. If the one or more additional data messages are detected by the scan, the EMA system **705** indicates that the first event is complete.

[0098] In some embodiments, the first event includes a maximum duration. A current duration of the first event is based on a time when the first data message was transmitted and the current time. The first event is considered expired when the current duration exceeds the maximum duration. In some embodiments, the maximum duration of the first event is greater than one day or a week, such as 14 days for a renewal as described above. In other embodiments, the maximum duration for the event is any period of time required to transmit the data messages of that event. The maximum duration could be from a few seconds to several months or even years.

[0099] If the second data message is not detected by the scan and the current duration of the first event exceeds the maximum duration of the first event, the EMA system **705** can transmit an error message to a user computer device. The EMA system **705** can also

[0100] If the second data message is not detected by the scan and the current duration of the first event does not exceed the maximum duration of the first event, the EMA system **705** can store a notification that the first event is ongoing. If the current duration of the first event is within a predetermined amount of time from the maximum duration of the first event, such as if there is one day remaining until the event expires, the EMA system **705** can transmit a reminder notification to complete the first event to a user computer device. The reminder notification includes instructions to transmit the second data message.

[0101] In some embodiments, the DIV system **115** receives a data message for validation. The DIV system **115** validates the received data message. The DIV system **115** provides a notification of the validation of the data message. And the DIV system **115** stores the validated data message in the plurality of data messages, such as in the EMA database **710**.

[0102] In some embodiments, the EMA system **705** detects a third data message in the plurality of data messages, where the third data message is associated with a second event that includes the third data message and a fourth data message. The EMA system **705** scans the plurality of data messages for the fourth data message associated with the second event.

[0103] In still further embodiments, the EMA system **705** detects a plurality of first data messages associated with a plurality of events. For each event of the plurality of events, the EMA system **705** determines one or more additional data messages associated with the corresponding event. For each event of the plurality of events, the EMA system **705** scans the plurality of data messages for the corresponding one or more additional events. In some embodiments, the EMA system **705** scans **805** the plurality of data messages for first data messages from a plurality of events. For each first data message found, the EMA system **705** identifies an event and searches for the rest of the data messages associated with that event. If the EMA system **705** detects all of the data messages associated with an event, then the EMA system **705** marks that event as completed. For each uncompleted event, the EMA system **705** determines **820** if the event has expired. If the event has expired, then the EMA system **705** transmits **825** and error message. If the event has not expired, the EMA system **705** stores information about the current state of the event and waits until the next

time process **800** is executed.

[0104] In some embodiments, process **800** is executed on a periodic basis, such as once a day, once an hour, or once a week. In other embodiments, process **800** is being performed continuously and the plurality of data messages are updated each time a new data message is transmitted. In these embodiments, the new data messages are identified to determine if they start new events or continue existing events.

[0105] FIG. **9** illustrates an example configuration of user computer device **902** used in the system **100** (shown in FIG. **4**), in accordance with one example of the present disclosure. User computer device **902** is operated by a user **901**. The user computer device **902** can include, but is not limited to, the source system **105** and the destination systems **125-135** (shown in FIG. **1**). The user computer device **902** includes a processor **905** for executing instructions. In some examples, executable instructions are stored in a memory area **910**. The processor **905** can include one or more processing units (e.g., in a multi-core configuration). The memory area **910** is any device allowing information such as executable instructions and/or transaction data to be stored and retrieved. The memory area **910** can include one or more computer-readable media.

[0106] The user computer device **902** also includes at least one media output component **915** for presenting information to the user **901**. The media output component **915** is any component capable of conveying information to the user **901**. In some examples, the media output component **915** includes an output adapter (not shown) such as a video adapter and/or an audio adapter. An output adapter is operatively coupled to the processor **905** and operatively coupleable to an output device such as a display device (e.g., a cathode ray tube (CRT), liquid crystal display (LCD), light emitting diode (LED) display, or “electronic ink” display) or an audio output device (e.g., a speaker or headphones). In some examples, the media output component **915** is configured to present a graphical user interface (e.g., a web browser and/or a client application) to the user **901**. A graphical user interface can include, for example, an interface for viewing validation errors. In some examples, the user computer device **902** includes an input device **920** for receiving input from the user **901**. The user **901** can use the input device **920** to, without limitation, select a validation error to view. The input device **920** can include, for example, a keyboard, a pointing device, a mouse, a stylus, a touch sensitive panel (e.g., a touch pad or a touch screen), a gyroscope, an accelerometer, a position detector, a biometric input device, and/or an audio input device. A single component such as a touch screen can function as both an output device of the media output component **915** and the input device **920**.

[0107] The user computer device **902** can also include a communication interface **925**, communicatively coupled to a remote device such as the DIV system **115** (shown in FIG. **1**), one or more destination systems **125-135**, and the source system **105**. The communication interface **925** can include, for example, a wired or wireless network adapter and/or a wireless data transceiver for use with a mobile telecommunications network.

[0108] Stored in the memory area **910** are, for example, computer-readable instructions for providing a user interface to the user **901** via the media output component **915** and, optionally, receiving and processing input from the input device **920**. A user interface can include, among other possibilities, a web browser and/or a client application. Web browsers enable users, such as the user **901**, to display and interact with media and other information typically embedded on a web page or a website from the source system **105** or the DIV system **115**. A client application allows the user **901** to interact with, for example, the DIV system **115**. For example, instructions can be stored by a cloud service, and the output of the execution of the instructions sent to the media output component **915**.

[0109] The processor **905** executes computer-executable instructions for implementing aspects of the disclosure.

[0110] FIG. **10** illustrates an example configuration of a server computer device **1001** used in the system **100** (shown in FIG. **1**), in accordance with one example of the present disclosure. Server

computer device **1001** can include, but is not limited to, the source system **105**, the DIV system **115**, and the destination systems **125-135** (shown in FIG. 1). The server computer device **1001** also includes a processor **1005** for executing instructions. Instructions can be stored in a memory area **1010**. The processor **1005** can include one or more processing units (e.g., in a multi-core configuration).

[0111] The processor **1005** is operatively coupled to a communication interface **1015** such that the server computer device **1001** is capable of communicating with a remote device such as another server computer device **1001**, the source system **105**, the DIV system **115**, and the destination systems **125-135**. For example, the communication interface **1015** can transmit data messages to the destination systems **125-135** via the Internet, as illustrated in FIG. 1.

[0112] The processor **1005** can also be operatively coupled to a storage device **1034**. The storage device **1034** is any computer-operated hardware suitable for storing and/or retrieving data, such as, but not limited to, data associated with the information source database **110** and the rules database (both shown in FIG. 1). In some examples, the storage device **1034** is integrated in the server computer device **1001**. For example, the server computer device **1001** can include one or more hard disk drives as the storage device **1034**. In other examples, the storage device **1034** is external to the server computer device **1001** and can be accessed by a plurality of server computer devices **1001**. For example, the storage device **1034** can include a storage area network (SAN), a network attached storage (NAS) system, and/or multiple storage units such as hard disks and/or solid state disks in a redundant array of inexpensive disks (RAID) configuration.

[0113] In some examples, the processor **1005** is operatively coupled to the storage device **1034** via a storage interface **1020**. The storage interface **1020** is any component capable of providing the processor **1005** with access to the storage device **1034**. The storage interface **1020** can include, for example, an Advanced Technology Attachment (ATA) adapter, a Serial ATA (SATA) adapter, a Small Computer System Interface (SCSI) adapter, a RAID controller, a SAN adapter, a network adapter, and/or any component providing the processor **1005** with access to the storage device **1034**.

[0114] The processor **1005** executes computer-executable instructions for implementing aspects of the disclosure. In some examples, the processor **1005** is transformed into a special purpose microprocessor by executing computer-executable instructions or by otherwise being programmed. For example, the processor **1005** is programmed with instructions such as those shown in FIGS. 5, 6, and 8.

[0115] Described herein are computer systems such as the data integrity and validation computer devices and related computer systems. As described herein, all such computer systems include a processor and a memory. However, any processor in a computer device referred to herein can also refer to one or more processors wherein the processor can be in one computing device or a plurality of computing devices acting in parallel. Additionally, any memory in a computer device referred to herein can also refer to one or more memories wherein the memories can be in one computing device or a plurality of computing devices acting in parallel.

[0116] As used herein, a processor can include any programmable system including systems using micro-controllers, reduced instruction set circuits (RISC), application-specific integrated circuits (ASICs), logic circuits, and any other circuit or processor capable of executing the functions described herein. The above examples are example only, and are thus not intended to limit in any way the definition and/or meaning of the term “processor.”

[0117] As used herein, the term “database” can refer to either a body of data, a relational database management system (RDBMS), or to both. As used herein, a database can include any collection of data including hierarchical databases, relational databases, flat file databases, object-relational databases, object oriented databases, and any other structured collection of records or data that is stored in a computer system. The above examples are example only, and thus are not intended to limit in any way the definition and/or meaning of the term database. Examples of RDBMS' include,

but are not limited to including, Oracle® Database, MySQL, IBM® DB2, Microsoft® SQL Server, Sybase®, and PostgreSQL. However, any database can be used that enables the systems and methods described herein. (Oracle is a registered trademark of Oracle Corporation, Redwood Shores, California; IBM is a registered trademark of International Business Machines Corporation, Armonk, New York; Microsoft is a registered trademark of Microsoft Corporation, Redmond, Washington; and Sybase is a registered trademark of Sybase, Dublin, California.)

[0118] In another example, a computer program is provided, and the program is embodied on a computer-readable medium. In an example, the system is executed on a single computer system, without requiring a connection to a server computer. In a further example, the system is being run in a Windows® environment (Windows is a registered trademark of Microsoft Corporation, Redmond, Washington). In yet another example, the system is run on a mainframe environment and a UNIX® server environment (UNIX is a registered trademark of X/Open Company Limited located in Reading, Berkshire, United Kingdom). In a further example, the system is run on an iOS® environment (iOS is a registered trademark of Cisco Systems, Inc. located in San Jose, CA). In yet a further example, the system is run on a Mac OS® environment (Mac OS is a registered trademark of Apple Inc. located in Cupertino, CA). In still yet a further embodiment, the system is run on Android® OS (Android is a registered trademark of Google, Inc. of Mountain View, CA). In another embodiment, the system is run on Linux® OS (Linux is a registered trademark of Linus Torvalds of Boston, MA). The application is flexible and designed to run in various different environments without compromising any major functionality.

[0119] In some examples, the system includes multiple components distributed among a plurality of computer devices. One or more components can be in the form of computer-executable instructions embodied in a computer-readable medium. The systems and processes are not limited to the specific embodiments described herein. In addition, components of each system and each process can be practiced independent and separate from other components and processes described herein. Each component and process can also be used in combination with other assembly packages and processes. The present examples can enhance the functionality and functioning of computers and/or computer systems.

[0120] As used herein, an element or step recited in the singular and proceeded with the word “a” or “an” should be understood as not excluding plural elements or steps, unless such exclusion is explicitly recited. Furthermore, references to “example” or “one example” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features.

[0121] As used herein, the terms “software” and “firmware” are interchangeable, and include any computer program stored in memory for execution by a processor, including RAM memory, ROM memory, EPROM memory, EEPROM memory, and non-volatile RAM (NVRAM) memory. The above memory types are example only, and are thus not limiting as to the types of memory usable for storage of a computer program.

[0122] Furthermore, as used herein, the term “real-time” refers to at least one of the time of occurrence of the associated events, the time of measurement and collection of predetermined data, the time to process the data, and the time of a system response to the events and the environment. In the examples described herein, these activities and events occur substantially instantaneously.

[0123] The systems and processes are not limited to the specific examples described herein. In addition, components of each system and each process can be practiced independent and separate from other components and processes described herein. Each component and process also can be used in combination with other assembly packages and processes.

[0124] The patent claims at the end of this document are not intended to be construed under 35 U.S.C. § 112(f) unless traditional means-plus-function language is expressly recited, such as “means for” or “step for” language being expressly recited in the claim(s).

[0125] The computer-implemented methods discussed herein can include additional, less, or

alternate actions, including those discussed elsewhere herein. The methods can be implemented via one or more local or remote processors, transceivers, servers, and/or sensors (such as processors, transceivers, servers, and/or sensors mounted on vehicles or mobile devices, or associated with smart infrastructure or remote servers), and/or via computer-executable instructions stored on non-transitory computer-readable media or medium. Additionally, the computer systems discussed herein can include additional, less, or alternate functionality, including that discussed elsewhere herein. The computer systems discussed herein may include or be implemented via computer-executable instructions stored on non-transitory computer-readable media or medium.

[0126] As used herein, the term “non-transitory computer-readable media” is intended to be representative of any tangible computer-based device implemented in any method or technology for short-term and long-term storage of information, such as, computer-readable instructions, data structures, program modules and sub-modules, or other data in any device. Therefore, the methods described herein can be encoded as executable instructions embodied in a tangible, non-transitory, computer readable medium, including, without limitation, a storage device and/or a memory device. Such instructions, when executed by a processor, cause the processor to perform at least a portion of the methods described herein. Moreover, as used herein, the term “non-transitory computer-readable media” includes all tangible, computer-readable media, including, without limitation, non-transitory computer storage devices, including, without limitation, volatile and nonvolatile media, and removable and non-removable media such as a firmware, physical and virtual storage, CD-ROMs, DVDs, and any other digital source such as a network or the Internet, as well as yet to be developed digital means, with the sole exception being a transitory, propagating signal.

[0127] This written description uses examples to disclose various implementations, including the best mode, and also to enable any person skilled in the art to practice the various implementations, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and can include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

Claims

1. A computer system for performing over-time event monitoring, the computer system including at least one processor in communication with at least one memory device, wherein the at least one processor is programmed to: scan a plurality of data messages transmitted over a computer network for one or more indicators of a first event, the one or more indicators including at least one of a message type, a destination system, a source system, or one or more payload fields of the plurality of data messages; identify a first data message, a second data message, and a third data message from the plurality of data messages as including an identifier associated with the first event, the first data message, the second data message, and the third data message being associated with first time data, second time data, and third time data respectively; determine that the second time data and the first time data indicate that the second data message was transmitted within a first predefined period of time from when the first data message was transmitted; determine that the third time data and the second time data indicate that the third data message was transmitted within a second predefined period of time from when the second data message was transmitted; and based upon the first predefined period of time and the second predefined period of time being satisfied, mark the first event as being complete in the at least one memory device.

2. The computer system of claim 1, wherein the at least one processor is further programmed to: determine that the first data message, the second data message, and the third data message were all

transmitted within a threshold period of time associated with a deadline; and mark the first event as being complete further based upon the deadline being satisfied.

3. The computer system of claim 1, wherein the at least one processor is further programmed to: identify the first data message and the second data message before identifying the third data message; determine that a deadline period of time associated with the first event is within a threshold amount of time from a current time and that the third data message has not yet been transmitted; and transmit an alert to a user computing device, the alert associated with the deadline period of time, the user computing device associated with the first event.

4. The computer system of claim 3, wherein the at least one processor is further programmed to identify the third data message after transmitting the alert.

5. The computer system of claim 1, wherein the first data message is associated with a start of the first event and the third data message is associated with an end of the first event.

6. The computer system of claim 5, wherein the at least one processor is further programmed to: identify the first data message before identifying the second data message and the third data message; and based upon identifying the first data message, further scan the plurality of data messages for the second data message and the third data message.

7. The computer system of claim 1, wherein the at least one processor is further programmed to: identify the first data message before identifying the second data message and the third data message; determine that at least one of the second data message or the third data message have not been transmitted and a threshold amount of time remaining before a deadline associated with the first event has passed; and transmit a reminder notification to a user computing device associated with the first event, the reminder notification associated with the at least one of the second data message or the third data message.

8. The computer system of claim 7, wherein the reminder notification includes instructions to transmit the at least one of the second data message or the third data message.

9. At least one non-transitory computer-readable storage medium with instructions stored thereon for performing over-time event monitoring, wherein the instructions, when executed by at least one processor, cause the at least one processor to: scan a plurality of data messages transmitted over a computer network for one or more indicators of a first event, the one or more indicators including at least one of a message type, a destination system, a source system, or one or more payload fields of the plurality of data messages; identify a first data message, a second data message, and a third data message from the plurality of data messages as including an identifier associated with the first event, the first data message, the second data message, and the third data message being associated with first time data, second time data, and third time data respectively; determine that the second time data and the first time data indicate that the second data message was transmitted within a first predefined period of time from when the first data message was transmitted; determine that the third time data and the second time data indicate that the third data message was transmitted within a second predefined period of time from when the second data message was transmitted; and based upon the first predefined period of time and the second predefined period of time being satisfied, mark the first event as being complete in the at least one non-transitory computer-readable storage medium.

10. The at least one non-transitory computer-readable storage medium of claim 9, wherein the instructions further cause the at least one processor to: determine that the first data message, the second data message, and the third data message were all transmitted within a threshold period of time associated with a deadline; and mark the first event as being complete further based upon the deadline being satisfied.

11. The at least one non-transitory computer-readable storage medium of claim 9, wherein the instructions further cause the at least one processor to: identify the first data message and the second data message before identifying the third data message; determine that a deadline period of time associated with the first event is within a threshold amount of time from a current time and

that the third data message has not yet been transmitted; and transmit an alert to a user computing device, the alert associated with the deadline period of time, the user computing device associated with the first event.

12. The at least one non-transitory computer-readable storage medium of claim 11, wherein the instructions further cause the at least one processor to identify the third data message after transmitting the alert.

13. The at least one non-transitory computer-readable storage medium of claim 9, wherein the first data message is associated with a start of the first event and the third data message is associated with an end of the first event.

14. The at least one non-transitory computer-readable storage medium of claim 13, wherein the instructions further cause the at least one processor to: identify the first data message before identifying the second data message and the third data message; and based upon identifying the first data message, further scan the plurality of data messages for the second data message and the third data message.

15. The at least one non-transitory computer-readable storage medium of claim 9, wherein the instructions further cause the at least one processor to: identify the first data message before identifying the second data message and the third data message; determine that at least one of the second data message or the third data message have not been transmitted and a threshold amount of time remaining before a deadline associated with the first event has passed; and transmit a reminder notification to a user computing device associated with the first event, the reminder notification associated with the at least one of the second data message or the third data message.

16. The at least one non-transitory computer-readable storage medium of claim 15, wherein the reminder notification includes instructions to transmit the at least one of the second data message or the third data message.

17. A method for performing over-time event monitoring implemented by at least one processor in communication with at least one memory, the method comprising: scanning a plurality of data messages transmitted over a computer network for one or more indicators of a first event, the one or more indicators including at least one of a message type, a destination system, a source system, or one or more payload fields of the plurality of data messages; identifying a first data message, a second data message, and a third data message from the plurality of data messages as including an identifier associated with the first event, the first data message, the second data message, and the third data message being associated with first time data, second time data, and third time data respectively; determining that the second time data and the first time data indicate that the second data message was transmitted within a first predefined period of time from when the first data message was transmitted; determining that the third time data and the second time data indicate that the third data message was transmitted within a second predefined period of time from when the second data message was transmitted; and based upon the first predefined period of time and the second predefined period of time being satisfied, marking the first event as being complete in the at least one memory.

18. The method of claim 17, further comprising: determining that the first data message, the second data message, and the third data message were all transmitted within a threshold period of time associated with a deadline; and marking the first event as being complete further based upon the deadline being satisfied.

19. The method of claim 17, further comprising: identifying the first data message and the second data message before identifying the third data message; determining that a deadline period of time associated with the first event is within a threshold amount of time from a current time and that the third data message has not yet been transmitted; transmitting an alert to a user computing device, the alert associated with the deadline period of time, the user computing device associated with the first event; and identifying the third data message after transmitting the alert.

20. The method of claim 17, wherein the first data message is associated with a start of the first

event and the third data message is associated with an end of the first event, and wherein the method further comprises: identifying the first data message before identifying the second data message and the third data message; and based upon identifying the first data message, further scanning the plurality of data messages for the second data message and the third data message.
